

Intelligent IR based anti-piracy monitoring system with biometric authentication for authorized camera access

1. How do the cameras work?

The Intelligent IR-Based Anti-Piracy Monitoring System with Biometric Authentication for Authorized Camera Access uses two ESP32-CAM modules, an ESP32-S3 controller, servo motors, and an operator dashboard to provide continuous surveillance and detect unauthorized recording attempts. Each camera performs a specific function to improve the overall monitoring efficiency.

The system uses two ESP32-CAM modules, each performing a different task.

Camera 1 (Sentry Camera)

Camera 1 (Sentry Camera) is mounted on a servo motor and continuously rotates within a predefined angle, such as 0° to 180° . This sweeping motion allows it to monitor a large area and eliminate blind spots. It continuously captures images or video frames and transmits them to the ESP32-S3 controller through a wireless connection. The servo motor is controlled by PWM signals generated by the ESP32-S3, enabling smooth and continuous scanning of the surroundings.

- * Mounted on a servo motor.
- * Continuously rotates between predefined angles (for example, 0° – 180°).
- * Monitors the entire surveillance area.
- * Captures live images continuously and transmits them to the ESP32-S3 controller.
- * Ensures that there are no blind spots within the protected region.

Camera 2 (Detection Camera)

Camera 2 (Detection Camera) Is fixed in the direction of the sensitive or restricted area where piracy attempts are more likely to occur. Unlike Camera 1, it does not rotate continuously. Instead, it focuses on capturing high-quality images of the protected area. The captured frames are also transmitted to the ESP32-S3 for analysis. Camera 2 provides detailed monitoring of suspicious activities and supports accurate person detection.

- * Faces the restricted area where piracy is most likely to occur.
- * Continuously captures live video.

- * Sends image frames to the ESP32-S3.
- * Used specifically for detecting people or suspicious activity.

ESP32-S3 Controller

- The ESP32-S3 acts as the central processing unit of the system. It receives image frames from both ESP32-CAM modules and processes them using embedded software. The controller manages the synchronization of both cameras, controls the servo motor, and communicates with the monitoring software through Wi-Fi. It also controls the activation of the IR emitter and laser module whenever a suspicious event is detected.
- The system follows a coordinated workflow. Camera 1 continuously scans the environment, while Camera 2 closely monitors the restricted zone. The ESP32-S3 combines the information received from both cameras to improve surveillance reliability. The processed data is transmitted to the operator dashboard, where live video streams, camera status, detection confidence, and event logs are displayed. This enables the operator to monitor the protected area in real time.
- The use of two dedicated cameras significantly improves monitoring efficiency. One camera ensures continuous surveillance over a wide area, while the other provides focused monitoring of high-risk locations. This dual-camera approach increases detection accuracy and minimizes the chances of missing unauthorized activities. As a result, the system provides an intelligent and reliable solution for anti-piracy monitoring while maintaining continuous observation of the secured environment.

The ESP32-S3 acts as the central processing unit.

It performs:

- * Receiving images from both cameras.
- * Image processing.
- * Person detection.
- * Decision making.
- * Servo motor control.

- * IR and laser control.

- * Wi-Fi communication with the monitoring software.

2. Can the System Stop Piracy Using Software.

The software component of the *Intelligent IR-Based Anti-Piracy Monitoring System* is designed to automatically detect suspicious activities and initiate appropriate responses. However, it is important to understand that software alone cannot physically disable or permanently stop another person's camera or recording device. Instead, the software acts as the intelligent decision-making component that monitors the environment and activates deterrent mechanisms when necessary.

The software continuously receives image frames from both ESP32-CAM modules through the ESP32-S3 controller. These images are analyzed using image-processing techniques and a lightweight detection algorithm. The detection engine identifies the presence of a person or suspicious movement within the protected area. Each detected object is assigned a confidence score, which helps reduce false detections caused by lighting changes, shadows, or environmental disturbances.

Once the confidence score exceeds a predefined threshold, the software determines that an unauthorized activity may be occurring. The ESP32-S3 then executes the programmed response by sending control signals through its GPIO pins. These signals activate the IR emitter and laser module connected to the controller. The IR emitter and laser are intended to act as deterrents by interfering with or discouraging unauthorized recording attempts, depending on the application and environmental conditions.

At the same time, the software sends an alert to the operator dashboard through the Wi-Fi network. The dashboard displays the live camera feed, detection confidence, event status, and timestamp, allowing the operator to verify the situation immediately. The detected event is also recorded in the system log for future analysis and security documentation.

The software operates autonomously without requiring continuous human intervention. After initialization, the system continuously captures images, processes video frames, evaluates detection results, and activates the required hardware modules whenever necessary. This closed-loop operation enables rapid response to suspicious activities while maintaining uninterrupted surveillance.

Therefore, the software does not directly disable cameras. Instead, it performs continuous monitoring, intelligent detection, automated decision-making, event logging, and activation of deterrent devices. These capabilities together provide an effective anti-piracy monitoring solution that enhances security, reduces response time, and assists operators in protecting sensitive environments from unauthorized recording attempts.

3. How the Software Works

The software architecture of the Intelligent IR-Based Anti-Piracy Monitoring System is designed as a complete automation system that integrates image acquisition, image processing, detection, decision-making, hardware control, and monitoring. The software runs on the ESP32-S3 controller along with a Python-based operator dashboard.

When the system is powered on, the ESP32-S3 initializes all connected hardware components, including the two ESP32-CAM modules, servo motor, IR emitter, laser module, and Wi-Fi interface. After successful initialization, the controller establishes communication with both cameras and begins receiving live image frames. If biometric authentication is included, the system first verifies the authorized user's identity before granting access to system controls.

Camera 1 continuously rotates using the servo motor, capturing images from different angles. Camera 2 remains focused on the protected area and continuously captures high-quality frames. Both cameras transmit their captured images to the ESP32-S3 for processing.

The image-processing module prepares each frame for analysis by reducing noise and converting it into a suitable format. The processed frames are passed to the detection engine, where a lightweight computer vision algorithm analyzes the images to identify the

presence of people or suspicious activities. Each detection is assigned a confidence score based on the probability of a valid detection.

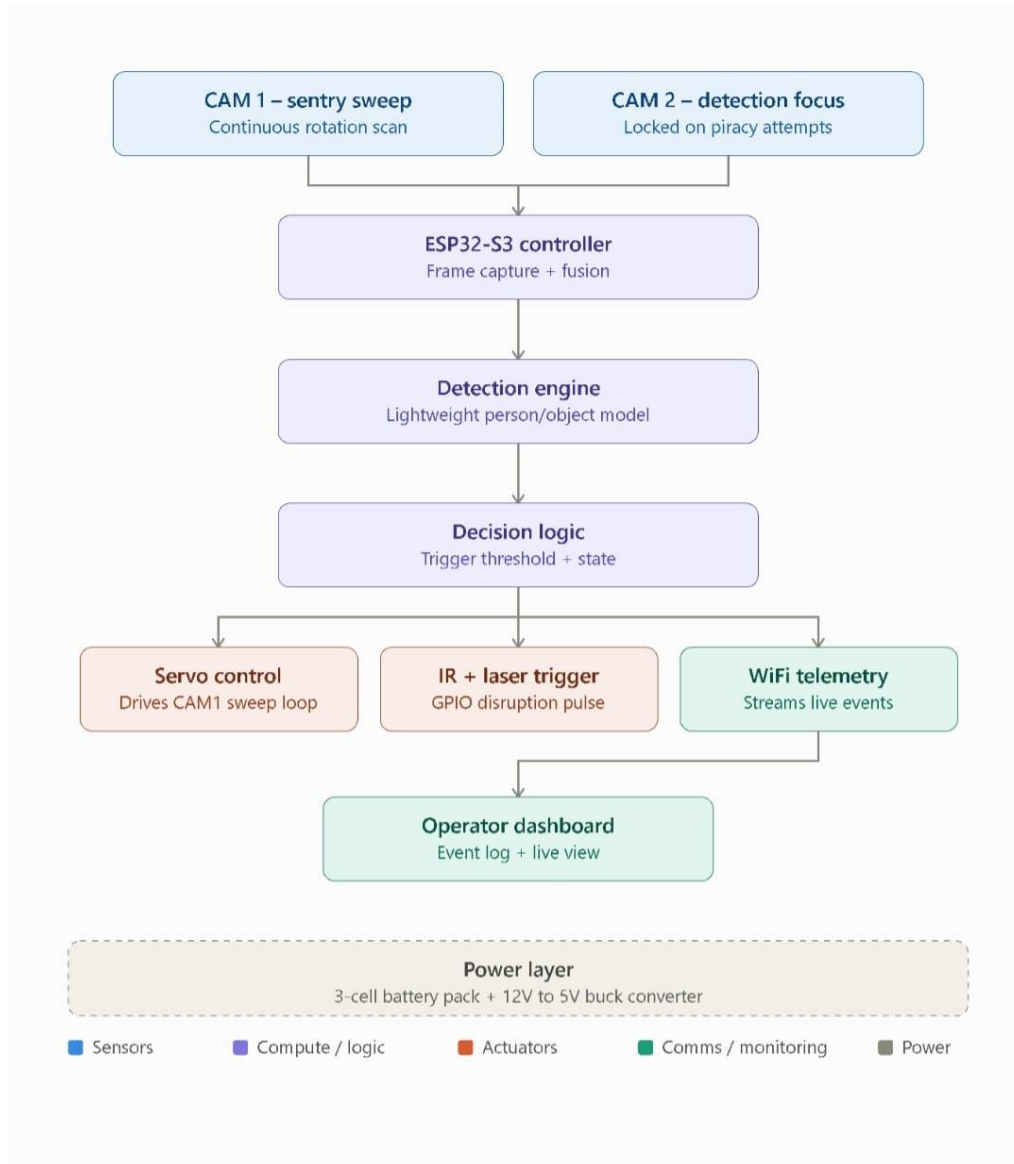
The decision logic compares the confidence score with a predefined threshold. If no suspicious activity is detected, the system continues monitoring normally while Camera 1 keeps scanning the environment. If the confidence score exceeds the threshold, the decision module immediately generates a detection event.

The ESP32-S3 then activates the IR emitter and laser module by generating GPIO control signals. Simultaneously, it transmits an alert to the operator dashboard over Wi-Fi. The dashboard displays the live video stream, detection status, confidence value, and recent event history. Every event is also stored in the system log with its corresponding date and time.

The software continues monitoring after each event and automatically returns to surveillance mode. Cool-down timers and confidence thresholds help minimize repeated triggers caused by temporary disturbances.

Overall, the software provides complete automation of surveillance, detection, hardware control, communication, and event logging. This integrated architecture enables continuous monitoring, faster response to unauthorized activities, and efficient coordination between all hardware components, making the system reliable and suitable for real-time anti-piracy applications.

4. Software Architecture



The software architecture of the Intelligent IR-Based Anti-Piracy Monitoring System with Biometric Authentication for Authorized Camera Access** is designed as a layered architecture that integrates image acquisition, biometric authentication, image processing, decision-making, hardware control, communication, and user monitoring. Each software module performs a specific function while communicating with other modules to provide intelligent, real-time surveillance and automated response.

The software execution begins with the ****System Initialization Module****. When power is supplied, the ESP32-S3 initializes all hardware components, including the two ESP32-CAM modules, servo motor, IR emitter, laser emitter, biometric sensor, Wi-Fi module, and GPIO interfaces. The system checks whether all devices are functioning correctly before entering monitoring mode. Network connectivity is established so that live data can be transmitted to the operator dashboard.

The next stage is the ****Biometric Authentication Module****. Before any authorized user can access the monitoring dashboard or modify system settings, the biometric sensor verifies the user's fingerprint or facial identity. If authentication is successful, the system grants access to the dashboard. If authentication fails, access is denied and the event is logged. This ensures that only authorized personnel can operate or configure the anti-piracy system.

After authentication, the ****Camera Acquisition Module**** becomes active. Camera 1 continuously rotates using the servo motor and captures images from different viewing angles, providing wide-area surveillance. Camera 2 continuously monitors the restricted area where piracy attempts are likely to occur. Both cameras capture image frames and transmit them to the ESP32-S3 through the wireless interface.

The captured frames are forwarded to the ****Image Processing Module****, where images are resized, filtered, and converted into a suitable format for analysis. Noise reduction and frame enhancement techniques improve image quality before further processing.

The processed images are then sent to the ****Detection Engine****. This module uses a lightweight computer vision algorithm to detect the presence of a person or suspicious activity. The detection engine calculates a confidence score for every detected object and compares it with a predefined threshold to reduce false detections caused by lighting changes or environmental noise.

The output of the detection engine is passed to the ****Decision Logic Module****. If no suspicious activity is detected, the system continues normal surveillance while Camera 1 keeps rotating. If the confidence score exceeds the threshold, the software identifies a potential piracy attempt and immediately initiates the response procedure.

The ****Hardware Control Module**** receives commands from the decision logic. It controls the servo motor, activates the IR emitter and laser emitter through GPIO signals, and maintains proper timing to avoid repeated triggering. The IR and laser modules act as deterrents to discourage unauthorized recording attempts while the system simultaneously records the event.

The ****Communication Module**** transmits live camera streams, system status, detection confidence, alerts, and event logs to the operator dashboard using Wi-Fi communication. The operator can monitor live video feeds, verify alerts, and review event history from the dashboard.

Finally, the ****Event Logging Module**** stores important information such as authentication attempts, detection events, timestamps, system status, and alerts for future analysis and security auditing.

This layered software architecture provides secure access through biometric authentication, continuous monitoring using dual ESP32-CAM modules, intelligent detection of suspicious activities, automated activation of deterrent devices, real-time communication with the operator dashboard, and comprehensive event logging, making it suitable for a robust anti-piracy monitoring system.

